QIntern 2026 Project Proposal

Project Title: Quantum Image Processing and Variational QML Mimar mîras for 3D Neuroimaging and Psychiatric Classification (Project Mr. Sina)

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Objectives

The primary objective of this project is to advance "Project Mr. Sina"—a clinical decision support system for chronic psychiatric disorders—by replacing classical neuroimaging pipelines and machine learning classifiers with a fully integrated Quantum Image Processing (QIP), Quantum Machine Learning (QML), and Quantum Optimization framework. The project leverages existing classical automated segmentation baselines (FSL/FreeSurfer) and upgrades them to model continuous-time brain alterations and volumetric features through quantum paradigms.

Specific objectives include:

- **Quantum Image Processing (QIP) Design:** Designing quantum state preparation algorithms (e.g., Flexible Representation of Quantum Images - FRQI or Novel Enhanced Quantum Representation - NEQR) adapted for 3D volumetric structural MRI data to enhance voxel-level contrast and reduce preprocessing noise.
- **Quantum Neural ODEs & QCNNs:** Implementing Quantum Convolutional Neural Networks (QCNNs) and continuous-time Quantum Neural Ordinary Differential Equations (Q-ODEs) to model structural brain variations (cortical thickness, subcortical volume trajectories) for high-accuracy psychiatric classification (Bipolar Disorder, Schizophrenia).
- **Quantum Optimization for Feature Selection:** Utilizing the Quantum Approximate Optimization Algorithm (QAOA) formulated as a Quadratic Unconstrained Binary Optimization (QUBO) problem to mathematically select the most relevant biomarkers from high-dimensional structural data arrays.
- **Hardware Execution & Error Mitigation:** Deploying the hybrid networks onto real quantum backends while incorporating error mitigation strategies to guarantee stability against quantum decoherence in clinical data processing.

Work Plan (1st July - 15 August)

Week 1 (July 1 - July 7): Technical Onboarding & QIP Formulation

- Map the existing classical NIfTI/FreeSurfer tabular outputs (aseg, aparc) into quantum data representation models.
- Review literature on 3D Quantum Image Processing (QIP) and quantum voxel-encoding strategies.

Week 2 (July 8 - July 14): Quantum State Preparation & Enhancement

- Develop quantum circuits for embedding structural MRI features into quantum states using Amplitude and Angle embedding.
- Design a prototype QIP filter for quantum-domain contrast enhancement of subcortical regions (e.g., Amygdala, Hippocampus).

Week 3 (July 15 - July 21): Hybrid QML Architecture Engineering

- Build a hybrid network combining classical feature extraction with Quantum Convolutional Neural Networks (QCNNs).
- Integrate continuous-time modeling principles by constructing a Parameterized Quantum Circuit (PQC) that functions within a Quantum Neural ODE framework to simulate longitudinal structural brain decay or alteration.

Week 4 (July 22 - July 28): Multi-Modal Quantum Optimization

- Formulate the high-dimensional brain biomarker selection problem into a QUBO model.
- Implement QAOA to optimize the choice of cortical thickness and subcortical volume parameters, maximizing diagnostic sensitivity while minimizing computational overhead.

Week 5 (July 29 - August 4): Real Hardware Evaluation & Noise Alleviation

- Run the optimized QIP filters and QML classifiers on simulated NISQ environments and deploy key circuits to real quantum processors (e.g., IBM Quantum Heron/Eagle backends).
- Apply Zero-Noise Extrapolation (ZNE) and Readout Error Mitigation to evaluate circuit performance under real-world quantum hardware noise constraints.

Week 6 (August 5 - August 15): Framework Integration & Documentation

- Combine the QIP, QML, and QAOA modules into a cohesive, reproducible codebase.
- Complete the open-source GitHub repository containing all source codes, Jupyter notebooks, and mathematical descriptions.
- Finalize the comprehensive research report and presentation slides for the QIntern closing assembly.

Deliverables

1. **Integrated Quantum-Medical Codebase:** A public GitHub repository containing functional modules for 3D Quantum Image Processing pipelines, QCNN/Q-ODE classifiers, and QAOA-based feature selection scripts.
2. **Educational Decks & Notebooks:** Step-by-step Jupyter notebooks illustrating how raw structural neuroimaging data maps into quantum amplitudes and how QUBO selects relevant psychiatric biomarkers.
3. **Comprehensive Academic Report:** A rigorous final document detailing the benchmarks of the quantum pipeline against classical baselines, highlighting achieved performance indicators, error mitigation outcomes, and future clinical deployment routes.